

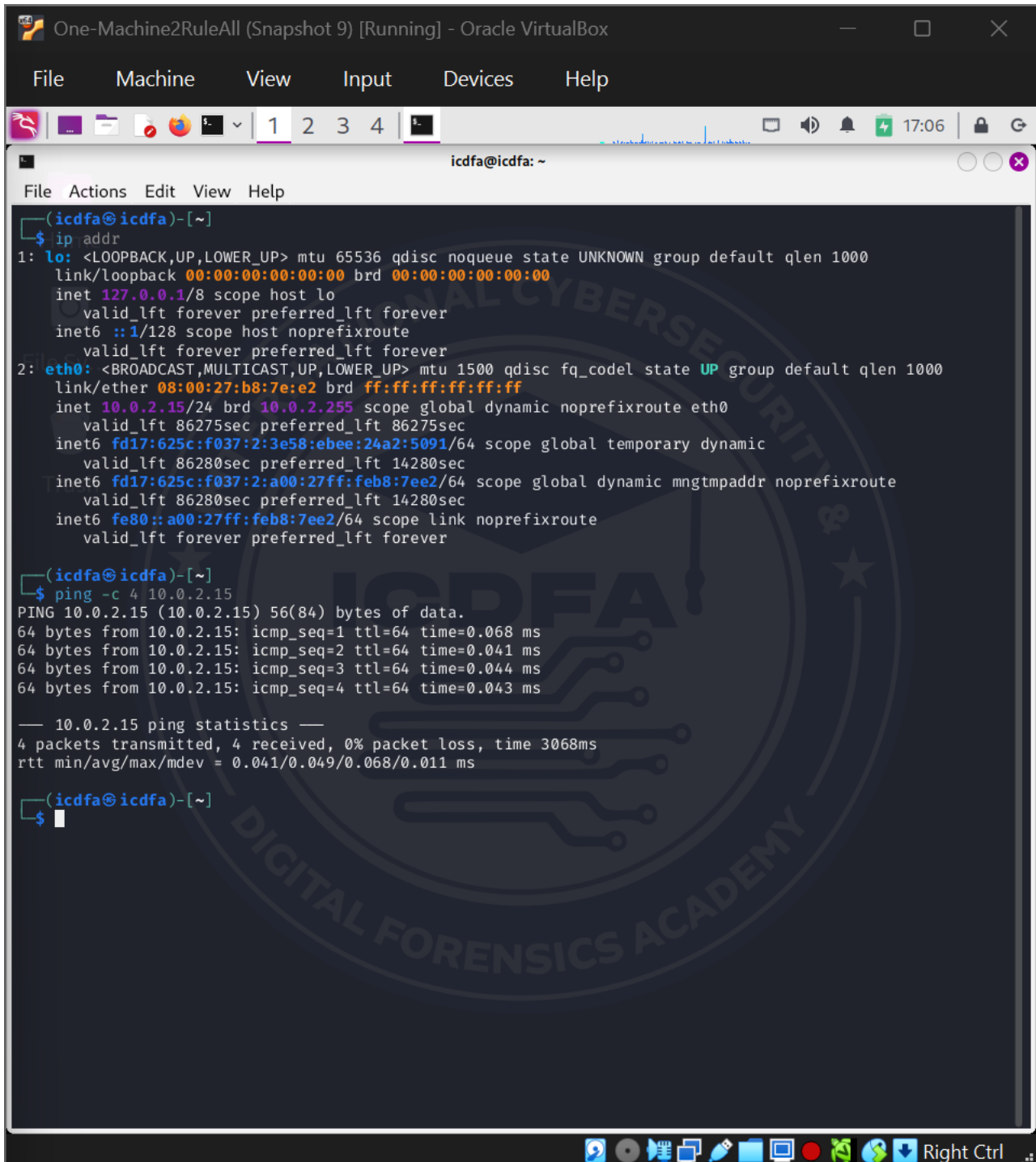
Full name: Adeleke Oluwapelumi Faruq

Registration number: C11/26/FCDF/17167

Linux distribution / version: Kali GNU/ 2025.1

VM hostname: Oracle VirtualBox

Part 1

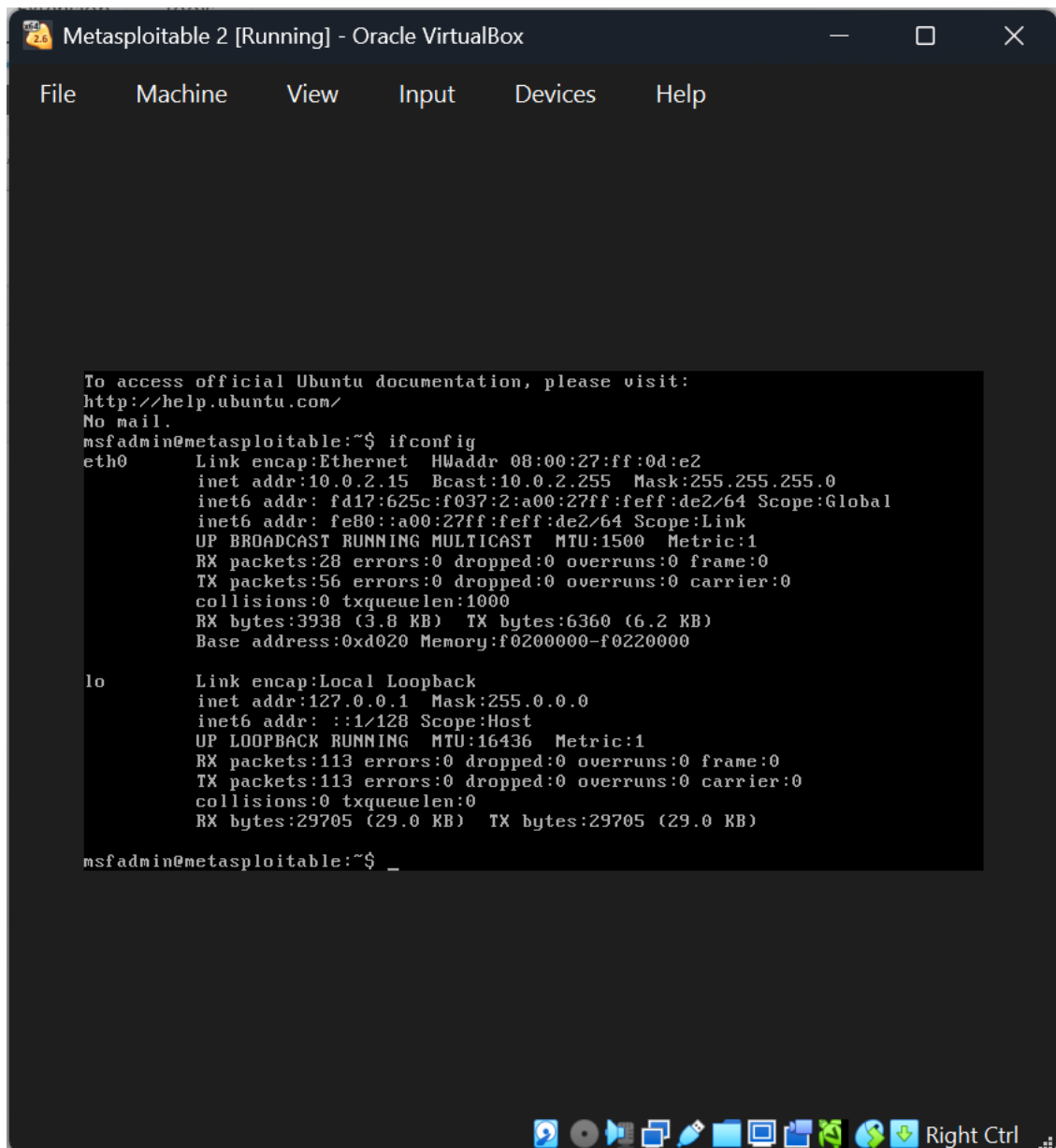


```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa@icdfa)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:b8:7e:e2 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86275sec preferred_lft 86275sec
   inet6 fd17:625c:f037:2:3e58:ebec:24a2:5091/64 scope global temporary dynamic
       valid_lft 86280sec preferred_lft 14280sec
   inet6 fd17:625c:f037:2:a00:27ff:feb8:7ee2/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86280sec preferred_lft 14280sec
   inet6 fe80::a00:27ff:feb8:7ee2/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

(icdfa@icdfa)-[~]
$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.068 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.041 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.044 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.043 ms

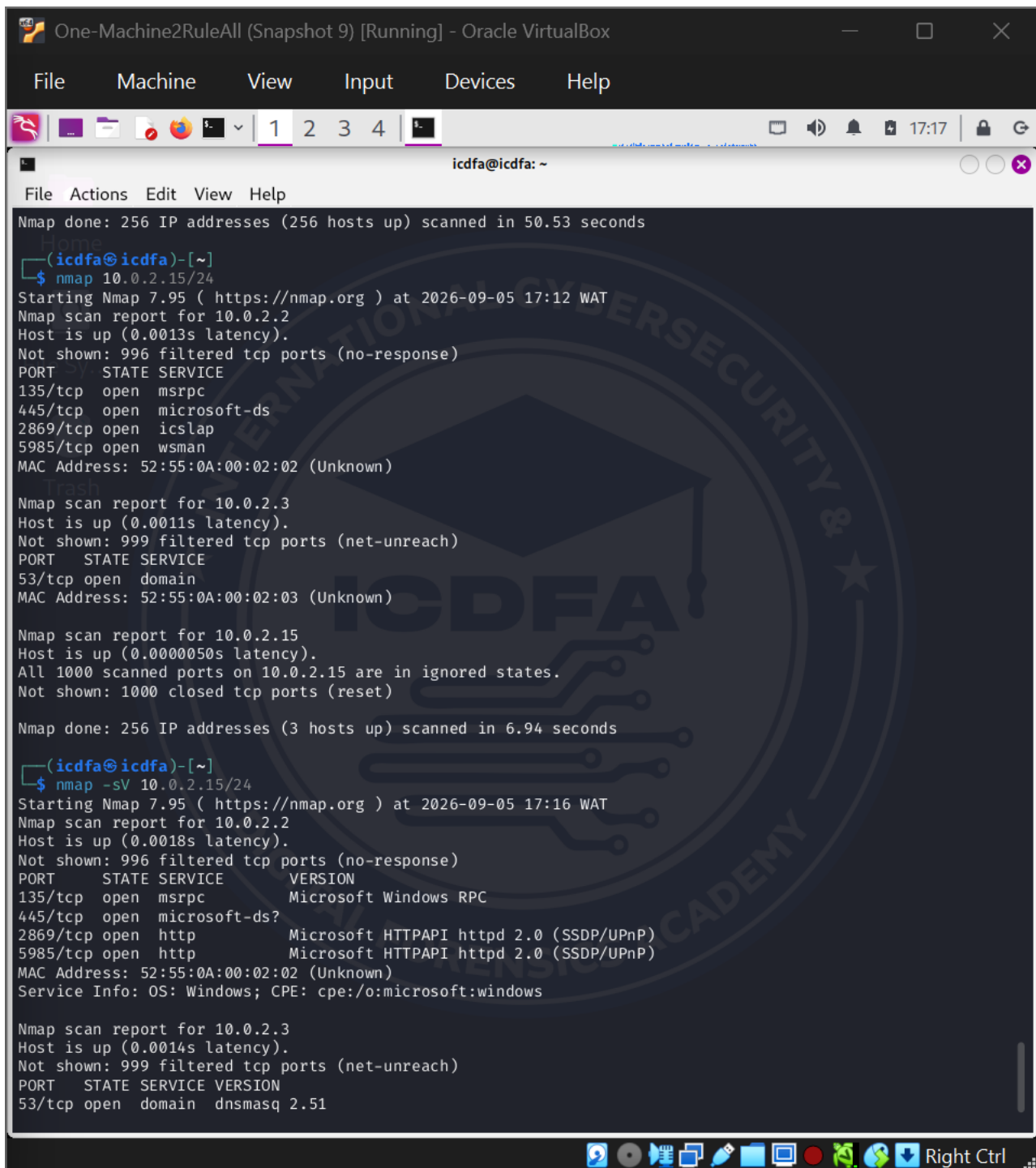
— 10.0.2.15 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3068ms
rtt min/avg/max/mdev = 0.041/0.049/0.068/0.011 ms

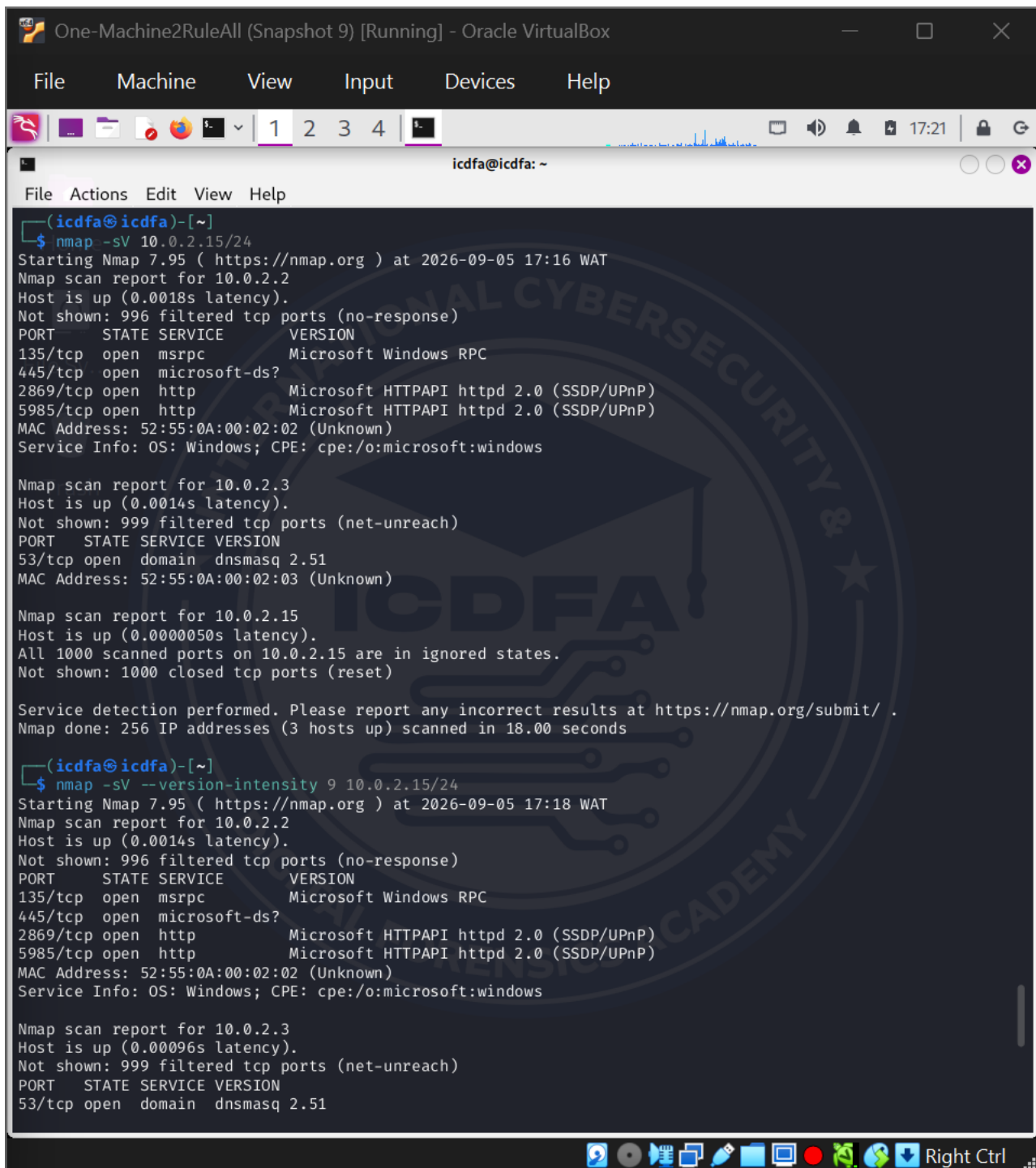
(icdfa@icdfa)-[~]
$
```



Part 3:

```
(icdfa@icdfa)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 17:09 WAT
Nmap scan report for 192.168.56.0
Host is up (0.00074s latency).
Nmap scan report for 192.168.56.1
Host is up (0.00071s latency).
Nmap scan report for 192.168.56.2
Host is up (0.00036s latency).
Nmap scan report for 192.168.56.3
Host is up (0.00043s latency).
Nmap scan report for 192.168.56.4
Host is up (0.00064s latency).
Nmap scan report for 192.168.56.5
Host is up (0.00066s latency).
Nmap scan report for 192.168.56.6
Host is up (0.00040s latency).
Nmap scan report for 192.168.56.7
Host is up (0.00027s latency).
Nmap scan report for 192.168.56.8
Host is up (0.00086s latency).
Nmap scan report for 192.168.56.9
Host is up (0.00063s latency).
Nmap scan report for 192.168.56.10
Host is up (0.00036s latency).
Nmap scan report for 192.168.56.11
Host is up (0.00065s latency).
Nmap scan report for 192.168.56.12
Host is up (0.0017s latency).
Nmap scan report for 192.168.56.13
Host is up (0.00032s latency).
Nmap scan report for 192.168.56.14
Host is up (0.00029s latency).
Nmap scan report for 192.168.56.15
Host is up (0.00039s latency).
Nmap scan report for 192.168.56.16
Host is up (0.00066s latency).
Nmap scan report for 192.168.56.17
Host is up (0.00035s latency).
Nmap scan report for 192.168.56.18
Host is up (0.00040s latency).
Nmap scan report for 192.168.56.19
Host is up (0.00030s latency).
Nmap scan report for 192.168.56.20
Host is up (0.00032s latency).
Nmap scan report for 192.168.56.21
Host is up (0.00030s latency).
```





```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help

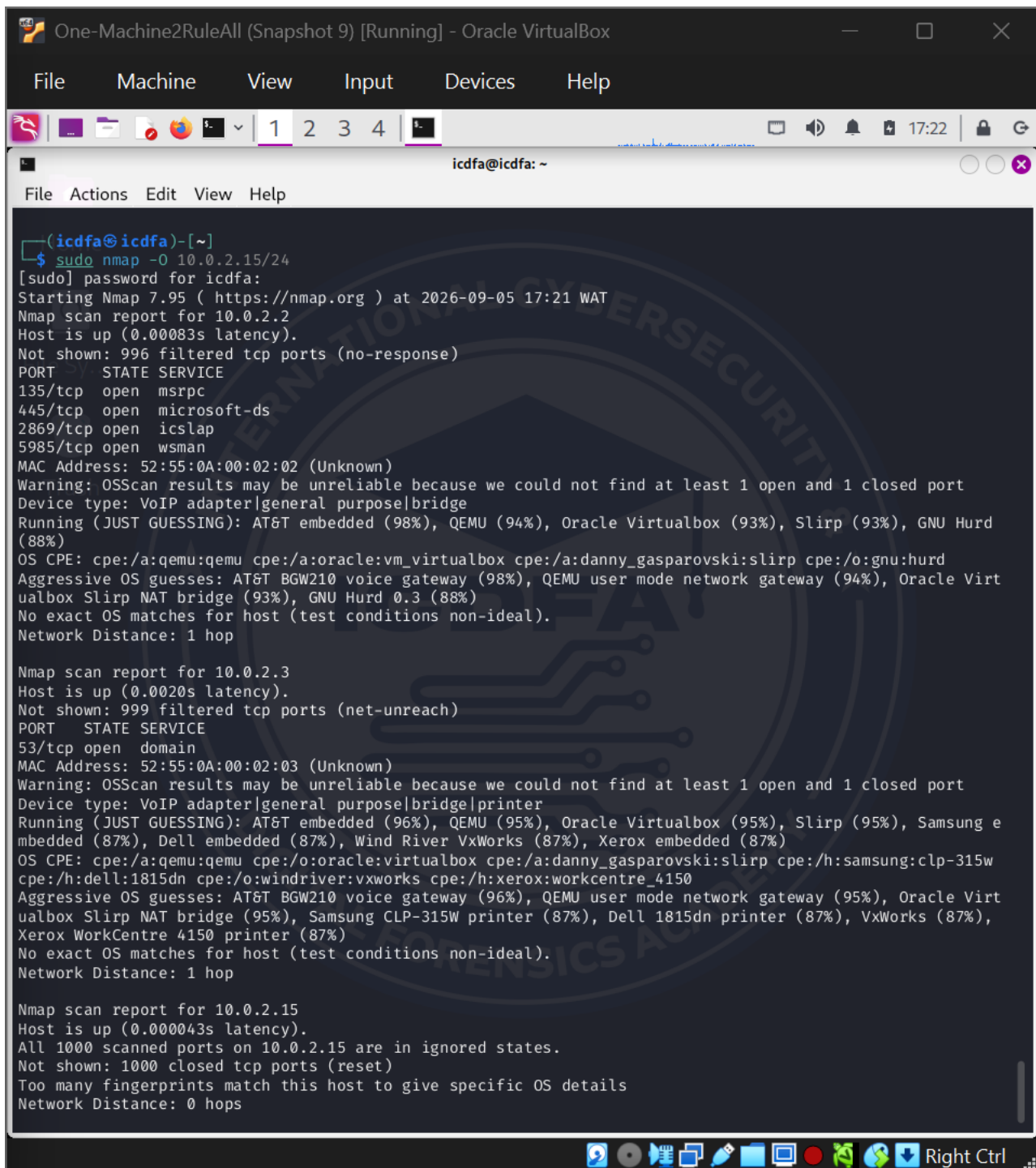
Nmap done: 256 IP addresses (3 hosts up) scanned in 18.00 seconds
(icdfa@icdfa)-[~]
$ nmap -sV --version-intensity 9 10.0.2.15/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 17:18 WAT
Nmap scan report for 10.0.2.2
Host is up (0.0014s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
445/tcp    open  microsoft-ds?
2869/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 52:55:0A:00:02:02 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

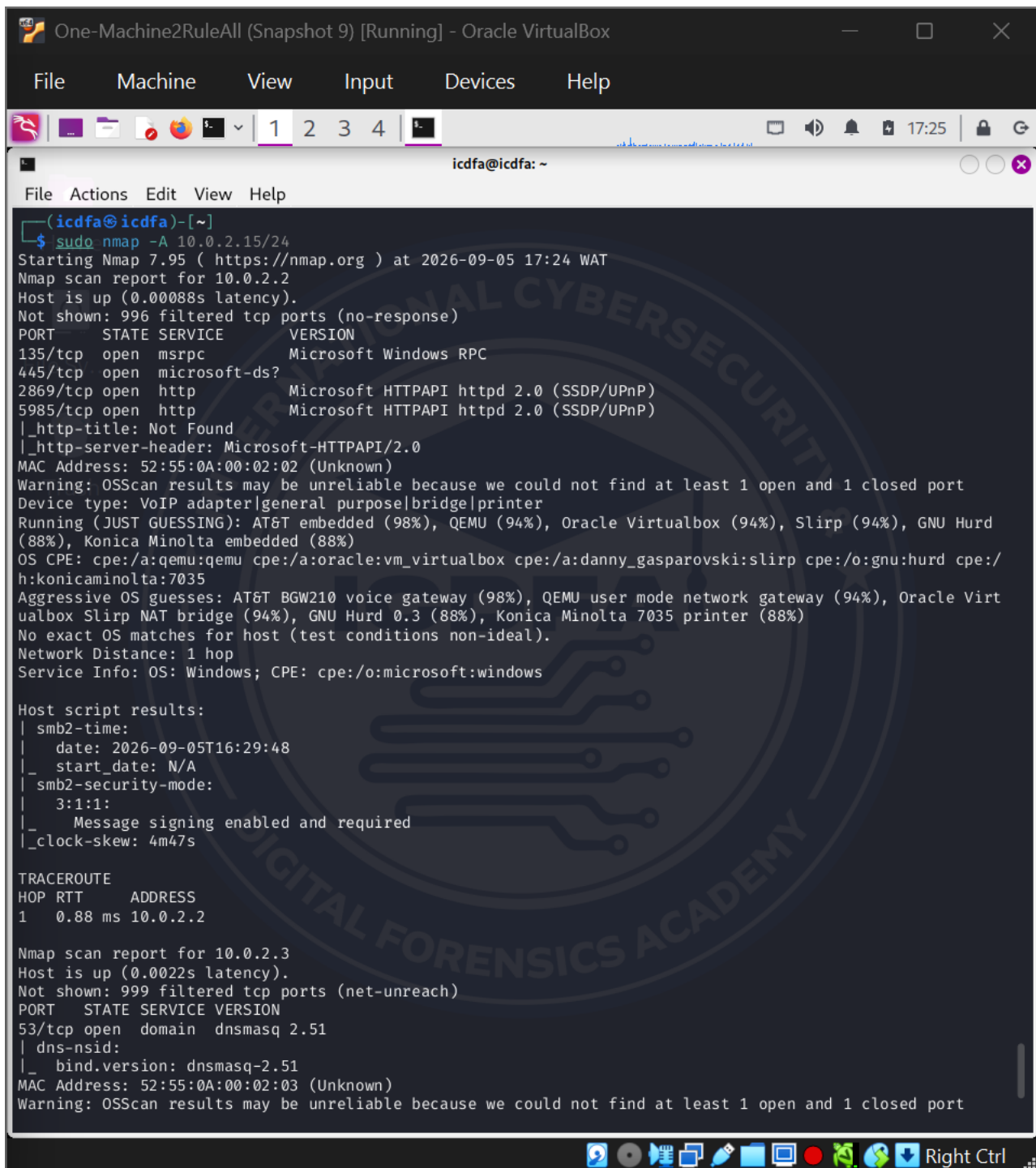
Nmap scan report for 10.0.2.3
Host is up (0.00096s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       dnsmasq 2.51
MAC Address: 52:55:0A:00:02:03 (Unknown)

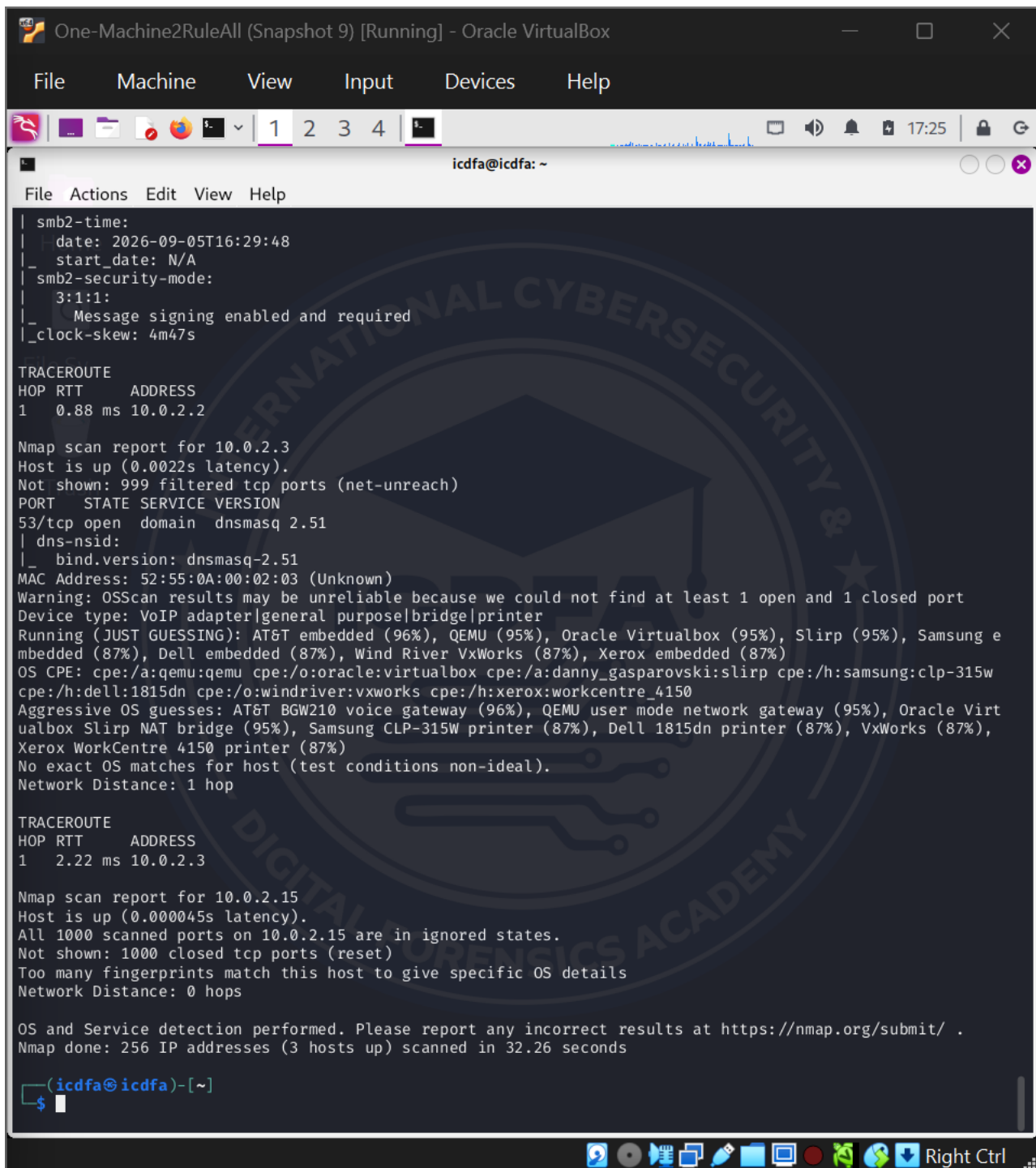
Nmap scan report for 10.0.2.15
Host is up (0.0000070s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

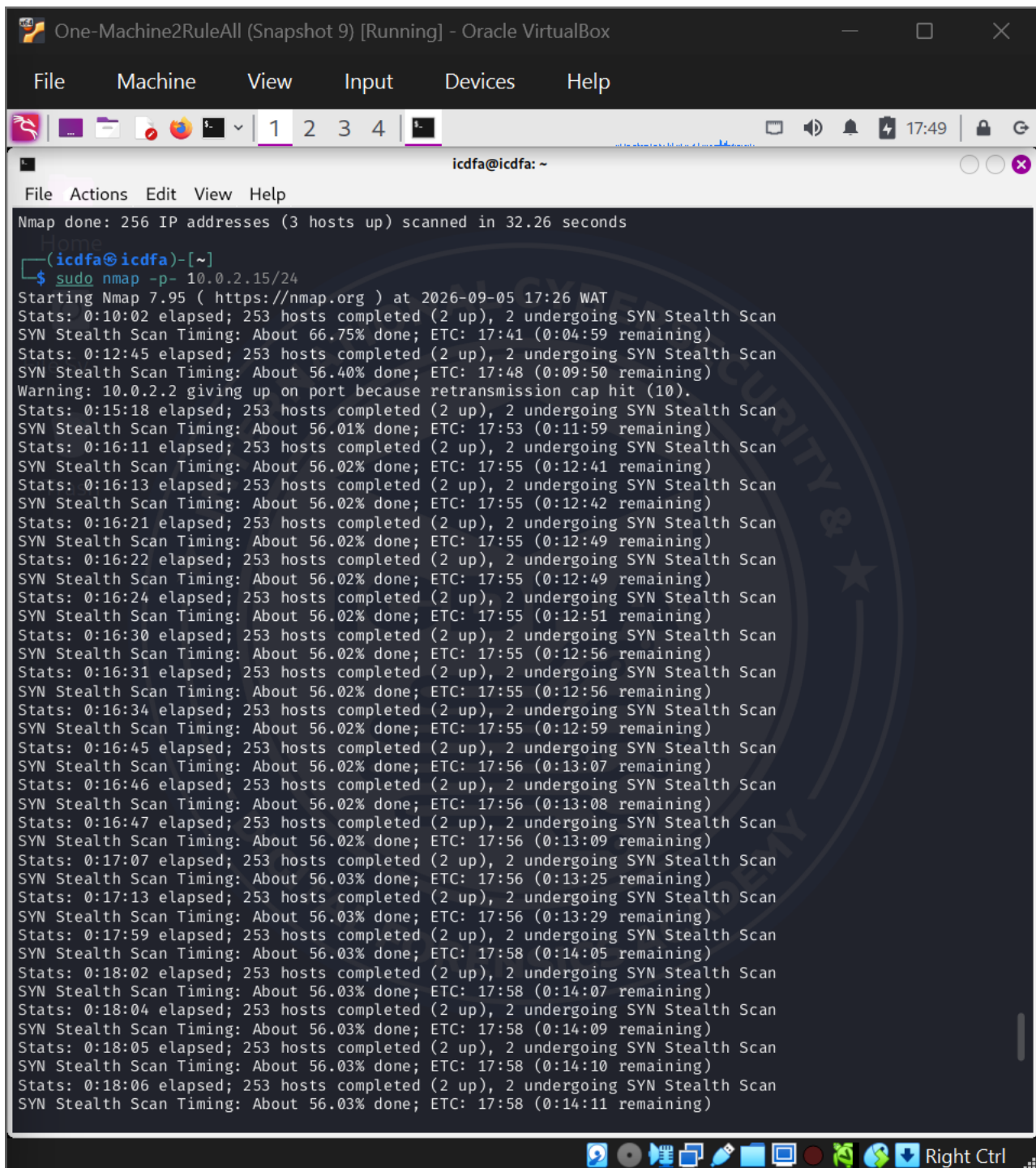
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 47.98 seconds

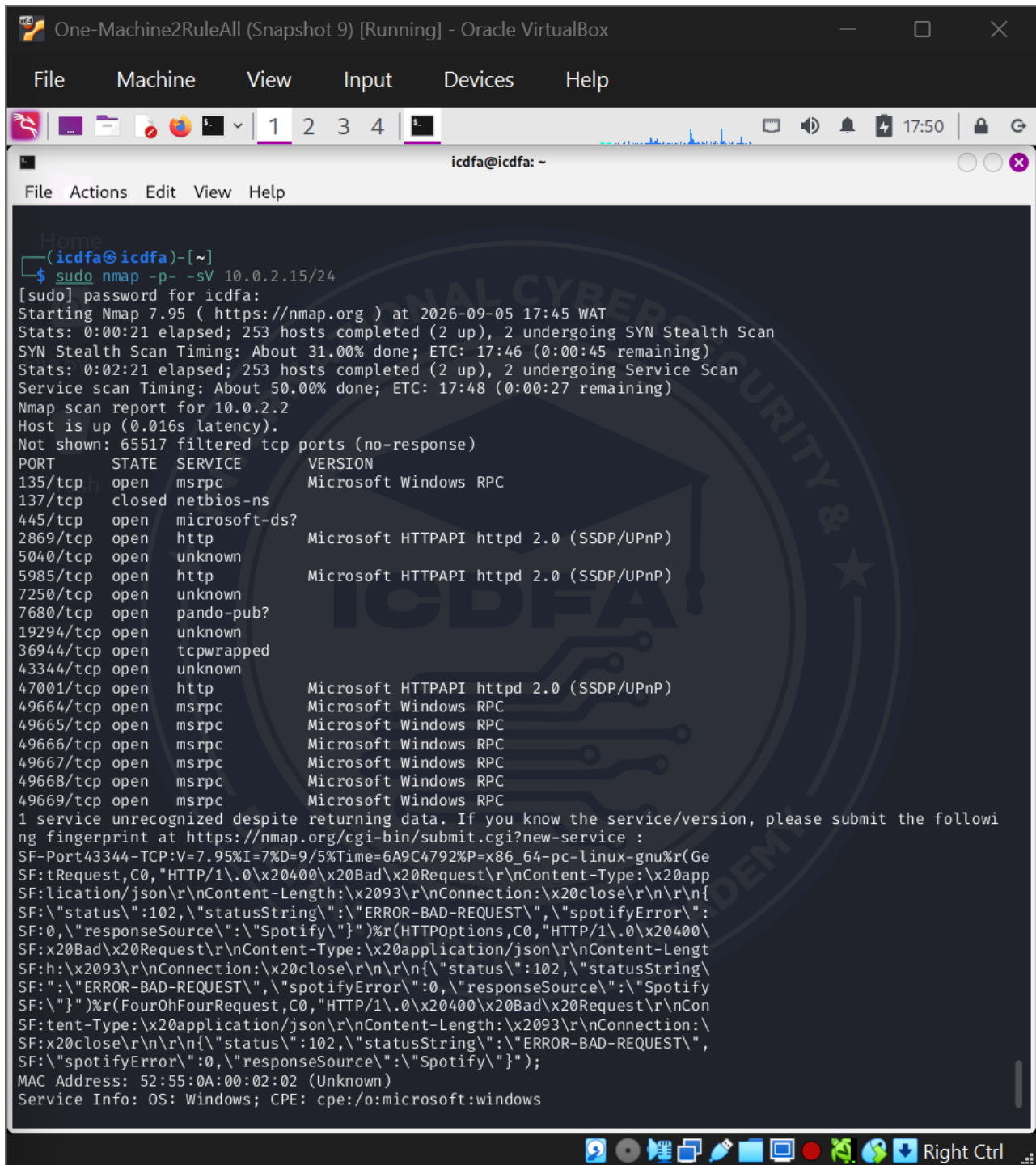
(icdfa@icdfa)-[~]
$ sudo nmap -O 10.0.2.15/24
[sudo] password for icdfa:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 17:21 WAT
Nmap scan report for 10.0.2.2
Host is up (0.00083s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
2869/tcp   open  icslap
5985/tcp   open  wsman
MAC Address: 52:55:0A:00:02:02 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|general purpose|bridge
Running (JUST GUESSING): AT&T embedded (98%), QEMU (94%), Oracle Virtualbox (93%), Slirp (93%), GNU Hurd (88%)
```

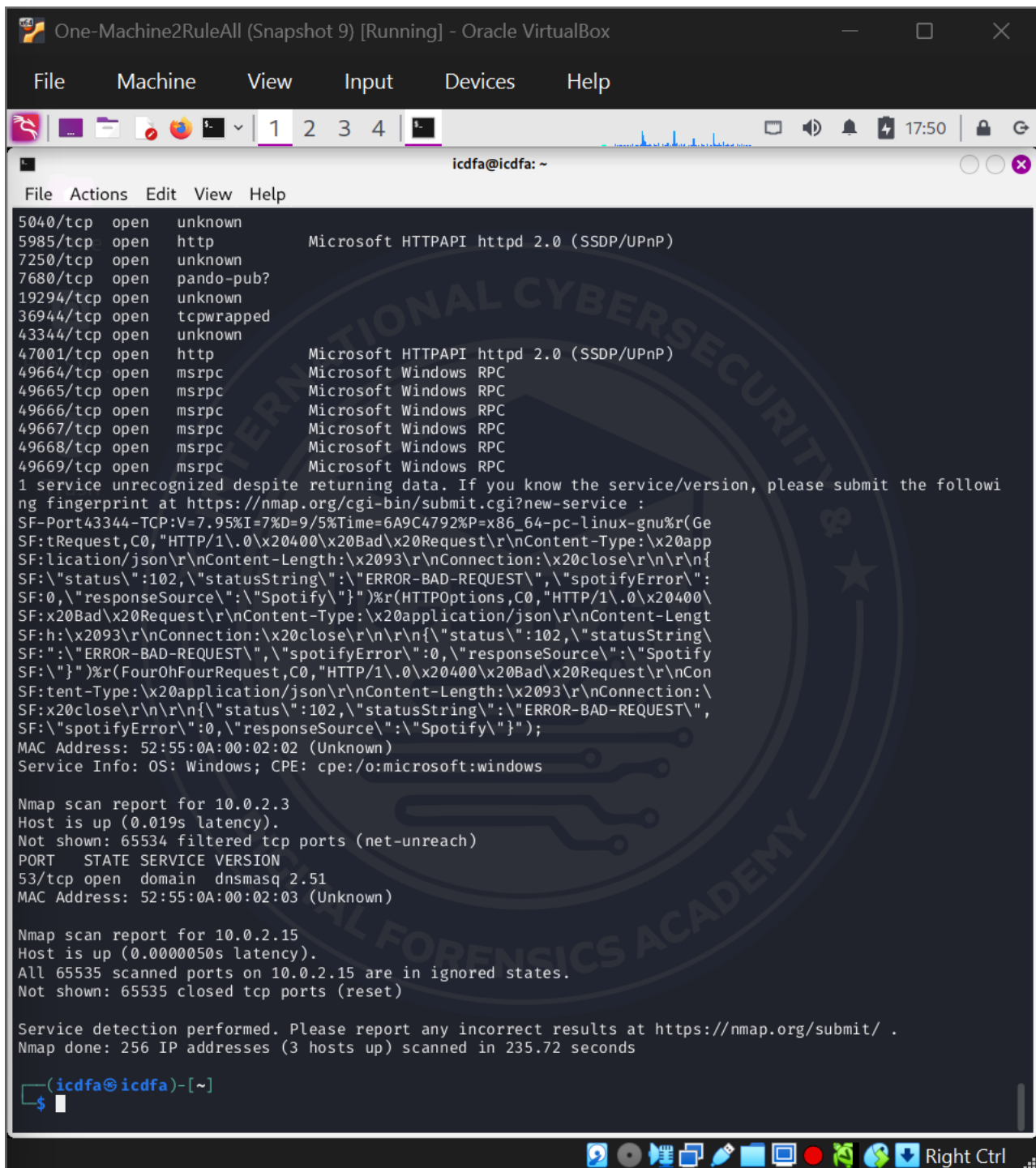




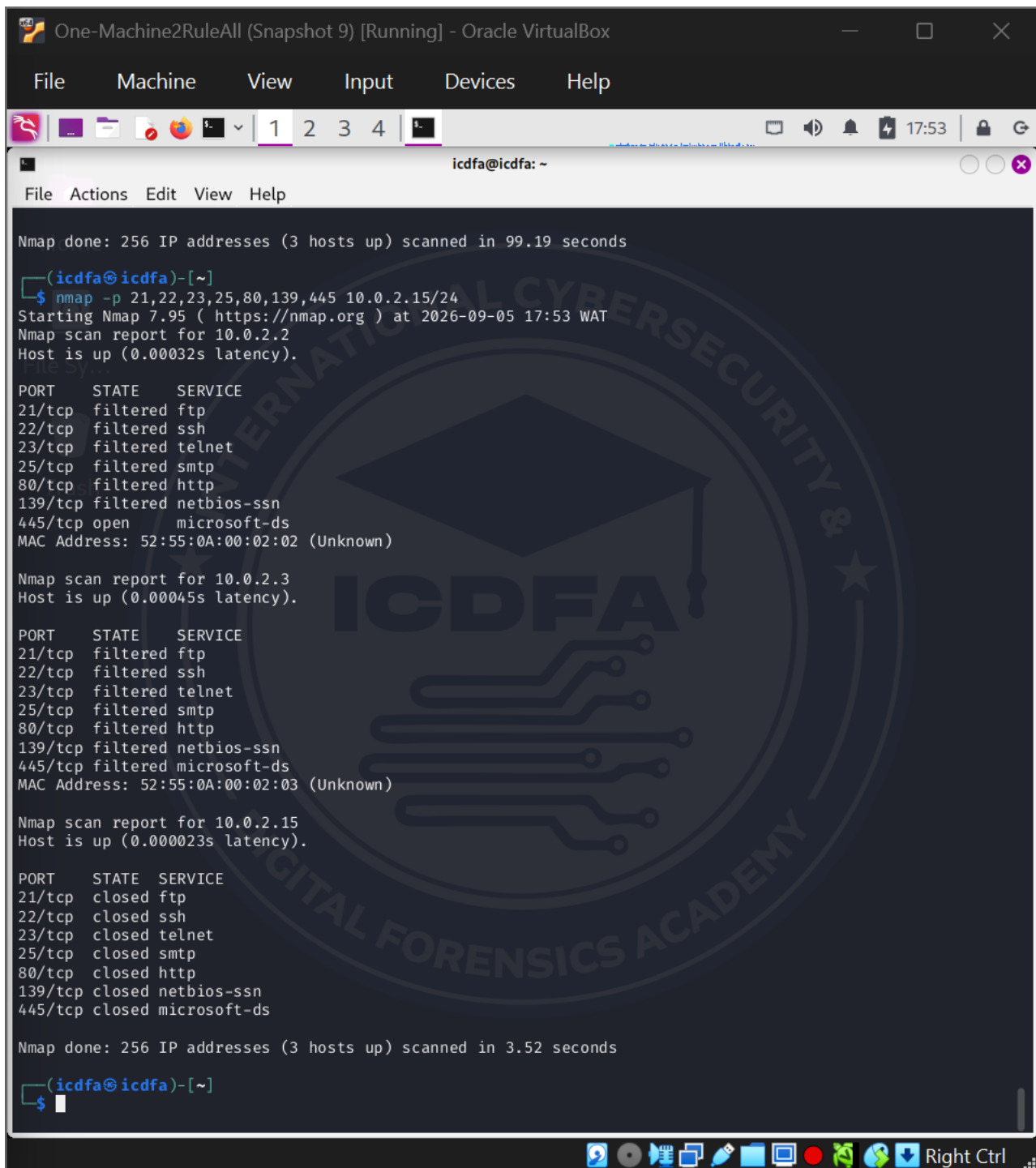


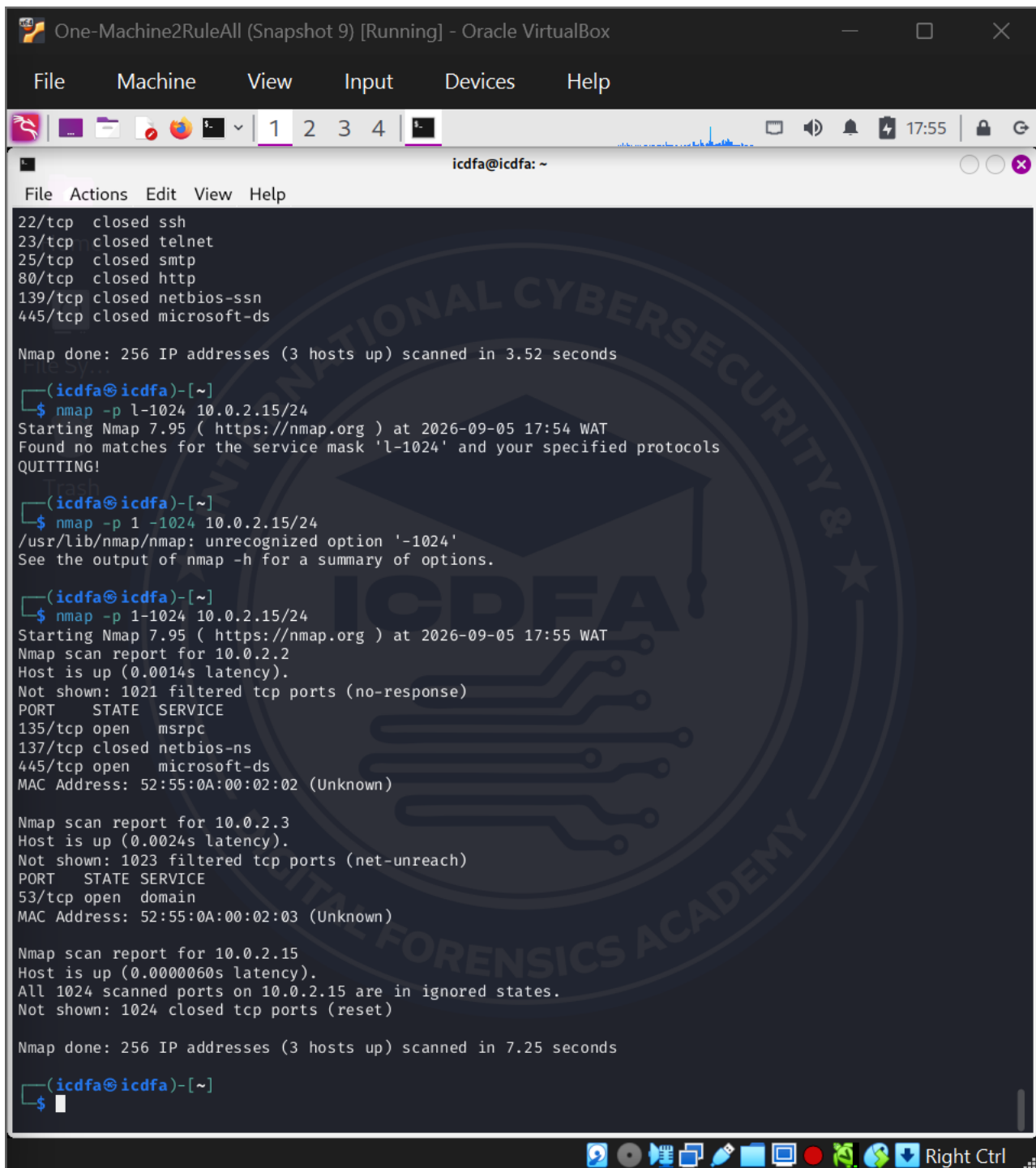


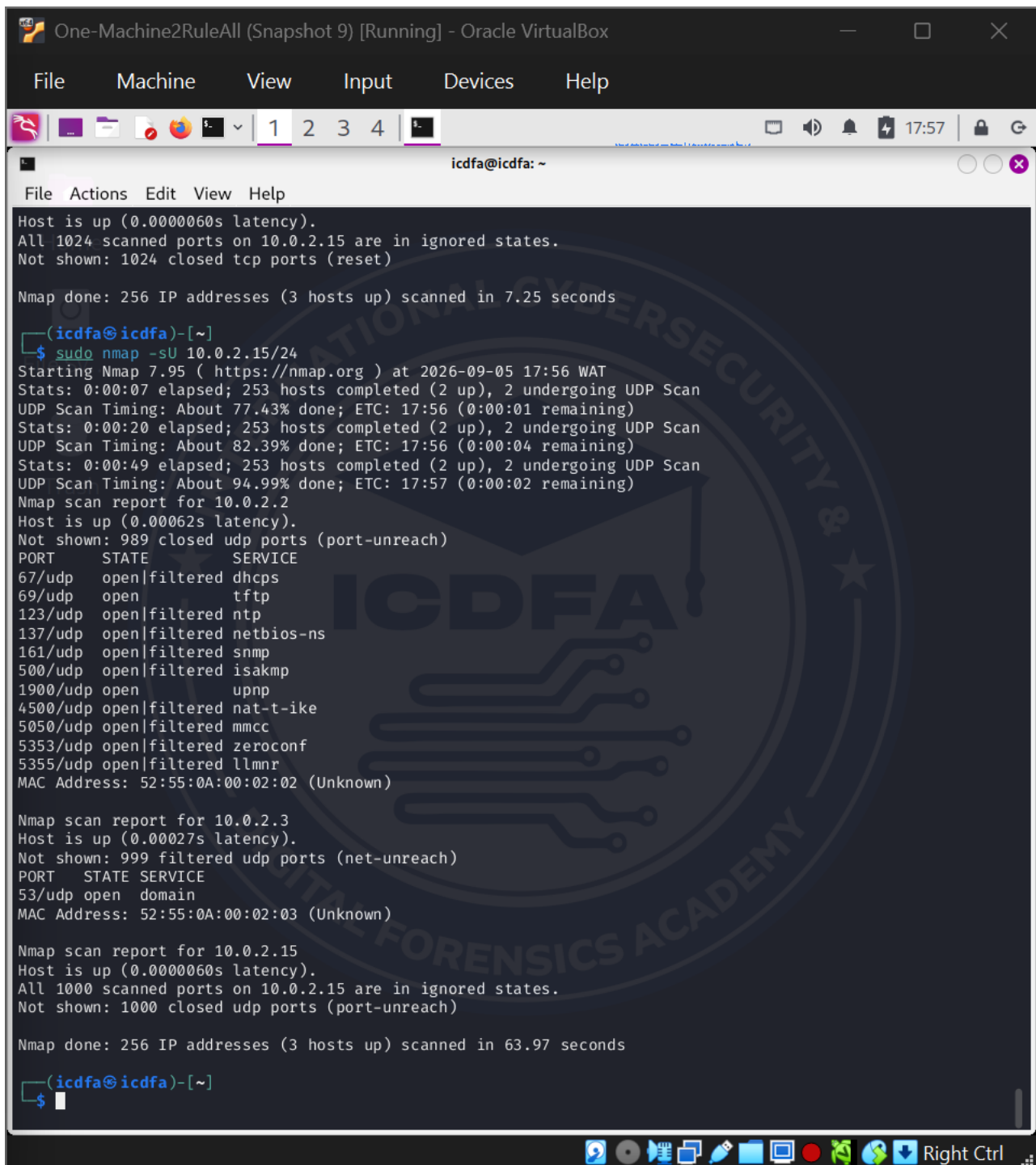












```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help

Nmap done: 256 IP addresses (3 hosts up) scanned in 63.97 seconds

(icdfa@icdfa)-[~]
$ sudo nmap -sU --top-ports 20 -sV 10.0.2.15/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 17:57 WAT
Stats: 0:00:22 elapsed; 253 hosts completed (2 up), 2 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 17:58 (0:00:30 remaining)
Stats: 0:01:50 elapsed; 253 hosts completed (2 up), 2 undergoing Service Scan
Service scan Timing: About 83.33% done; ETC: 18:00 (0:00:21 remaining)
Nmap scan report for 10.0.2.2
Host is up (0.039s latency).

PORT      STATE      SERVICE      VERSION
53/udp    closed     domain
67/udp    open|filtered dhcp
68/udp    closed     dhcp
69/udp    open       tftp
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp   closed     netbios-ns
138/udp   closed     netbios-dgm
139/udp   closed     netbios-ssn
161/udp   open|filtered snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   closed     route
631/udp   closed     ipp
1434/udp  closed     ms-sql-m
1900/udp  open       upnp?
4500/udp  open|filtered nat-t-ike
49152/udp closed     unknown

2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port69-UDP:V=7.95%I=7%D=9/5%Time=6A9C4A25%P=x86_64-pc-linux-gnu%r(xdmcp
SF:,15,"\\0\\x05\\0\\x02Access\\x20violation\\0");
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port1900-UDP:V=7.95%I=7%D=9/5%Time=6A9C4A20%P=x86_64-pc-linux-gnu%r(UPN
SF:P_MSEARCH,B2B,"HTTP/1\\.1\\x20200\\x200K\\r\\nST:\\x20urn:schemas-upnp-org:se
SF:rvice:ConnectionManager:1\\r\\nUSN:\\x20uuid:104cf802-dfd0-47bb-8a7c-09955
SF:59fa087::urn:schemas-upnp-org:service:ConnectionManager:1\\r\\nLocation:\\
SF:x20http://127\\.0\\.0\\.1:2869/upnphost/udhisapi\\.dll?content=uuid:104cf8
SF:02-dfd0-47bb-8a7c-0995559fa087\\r\\nOPT:\\x20http://schemas\\.upnp\\.org/upnp/
SF:1/0/\\";\\x20ns=01\\r\\n01-NLS:\\x20dd8a8940c3c20efa47c95c2ab3f296ff\\r\\nCach
SF:e-Control:\\x20max-age=1800\\r\\nServer:\\x20Microsoft-Windows/10\\.0\\x20UPn
```

One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

1234

18:00

icdfa@icdfa: ~

FileActionsEditViewHelp

SF:rvice:ConnectionManager:1\r\nUSN:\x20uuid:104cf802-dfd0-47bb-8a7c-0995559fa087::urn:schemas-upnp-org:service:ConnectionManager:1\r\nLocation:\x20http://127.0.0.1:2869/upnphost/udhisapi.dll?content=uuid:104cf802-dfd0-47bb-8a7c-0995559fa087\r\nOPT:\x20"http://schemas.upnp.org/upnp/1/0/";\x20ns=01\r\n01-NLS:\x20dd8a8940c3c20efa47c95c2ab3f296ff\r\nCacheControl:\x20max-age=1800\r\nServer:\x20Microsoft-Windows/10.0\x20UPnPDevice-Host/1.0\r\nExt:\x20\r\nDate:\x20Sat,\x2005\x20Sep\x202026\x2015:55:54\x20GMT\r\n\r\nHTTP/1.1\x20200\x20K\r\nST:\x20urn:schemas-upnp-org:service:RenderingControl:1\r\nUSN:\x20uuid:104cf802-dfd0-47bb-8a7c-0995559fa087::urn:schemas-upnp-org:service:RenderingControl:1\r\nLocation:\x20http://127.0.0.1:2869/upnphost/udhisapi.dll?content=uuid:104cf802-dfd0-47bb-8a7c-0995559fa087\r\nOPT:\x20"http://schemas.upnp.org/upnp/1/0/";\x20ns=01\r\n01-NLS:\x20dd8a8940c3c20efa47c95c2ab3f296ff\r\nCache-Control:\x20max-age=1800\r\nServer:\x20";\r\nMAC Address: 52:55:0A:00:02:02 (Unknown)

Nmap scan report for 10.0.2.3

Host is up (0.0034s latency).

PORT	STATE	SERVICE	VERSION
53/udp	open	domain	dnsmasq 2.51
67/udp	filtered	dhcps	
68/udp	filtered	dhcpc	
69/udp	filtered	tftp	
123/udp	filtered	ntp	
135/udp	filtered	msrpc	
137/udp	filtered	netbios-ns	
138/udp	filtered	netbios-dgm	
139/udp	filtered	netbios-ssn	
161/udp	filtered	snmp	
162/udp	filtered	snmptrap	
445/udp	filtered	microsoft-ds	
500/udp	filtered	isakmp	
514/udp	filtered	syslog	
520/udp	filtered	route	
631/udp	filtered	ipp	
1434/udp	filtered	ms-sql-m	
1900/udp	filtered	upnp	
4500/udp	filtered	nat-t-ike	
49152/udp	filtered	unknown	

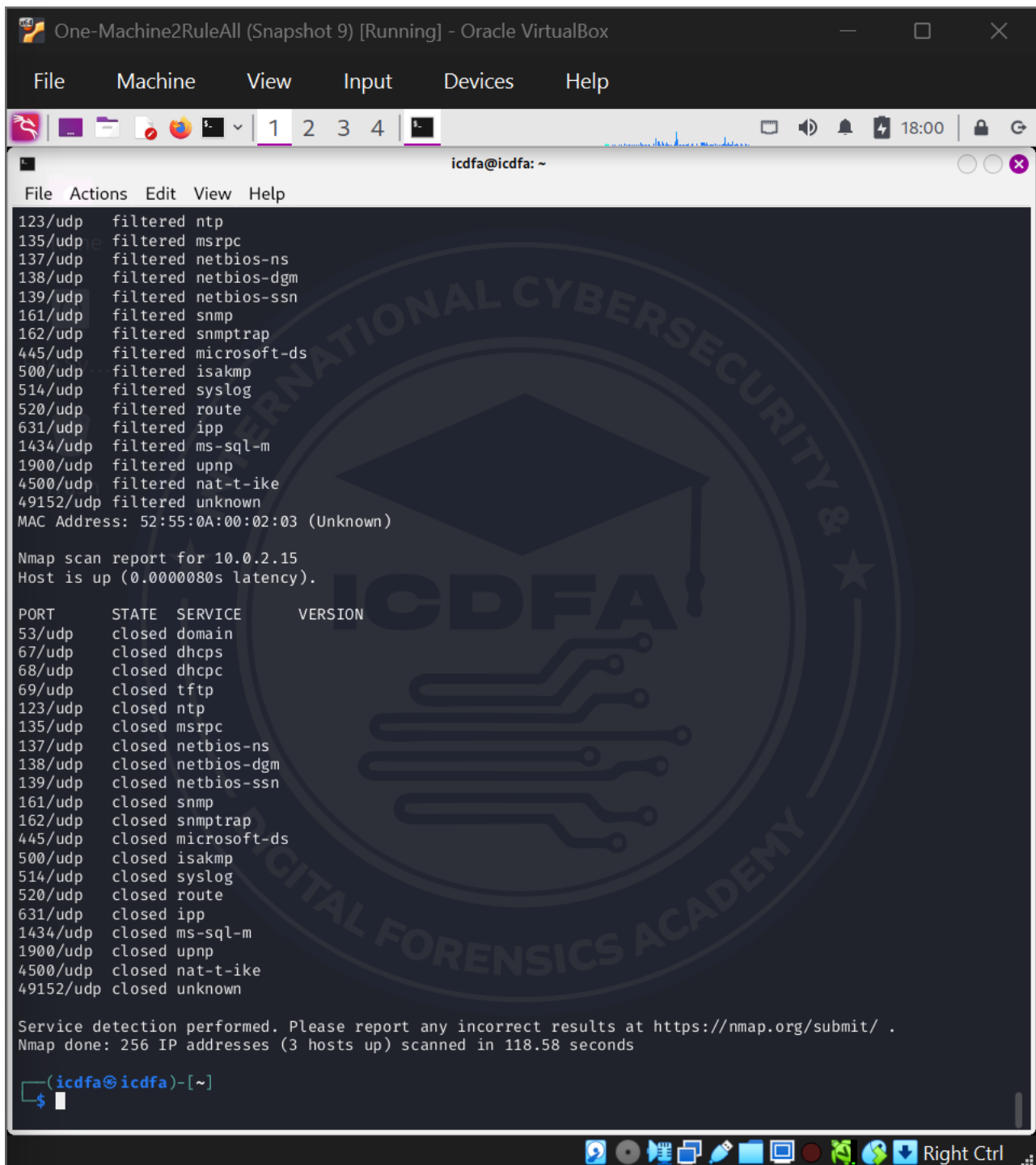
MAC Address: 52:55:0A:00:02:03 (Unknown)

Nmap scan report for 10.0.2.15

Host is up (0.0000080s latency).

PORT	STATE	SERVICE	VERSION
53/udp	closed	domain	
67/udp	closed	dhcps	

Right Ctrl



Part 4

```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help

Nmap done: 256 IP addresses (3 hosts up) scanned in 118.58 seconds
Home
(icdfa@icdfa)-[~]
$ nmap -sC -sV 10.0.2.15/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 18:02 WAT
Nmap scan report for 10.0.2.2
Host is up (0.0014s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc           Microsoft Windows RPC
445/tcp    open  microsoft-ds?
2869/tcp   open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp   open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Not Found
|_ http-server-header: Microsoft-HTTPAPI/2.0
MAC Address: 52:55:0A:00:02:02 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

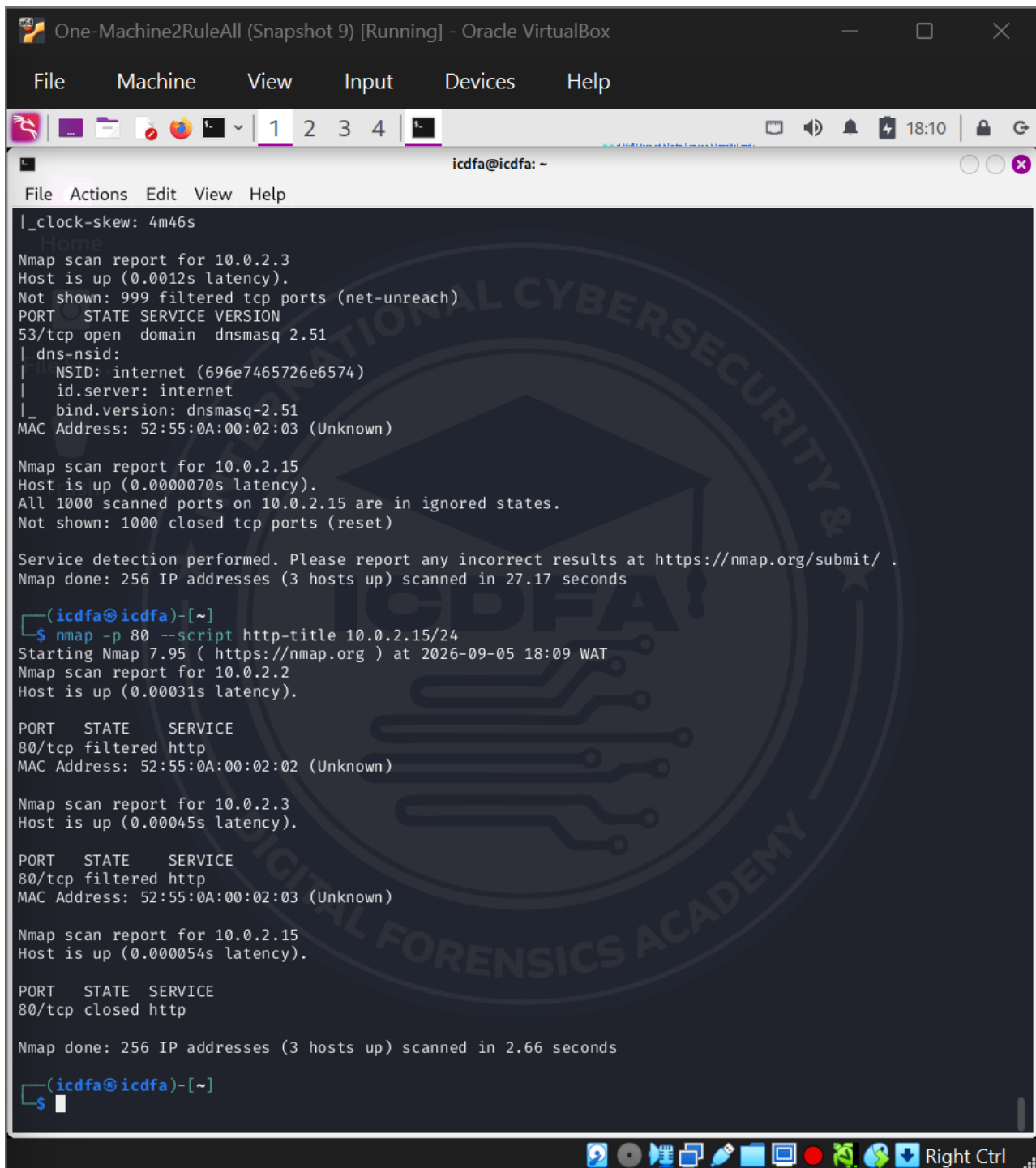
Host script results:
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-09-05T17:07:59
|_  start_date: N/A
|_ clock-skew: 4m46s

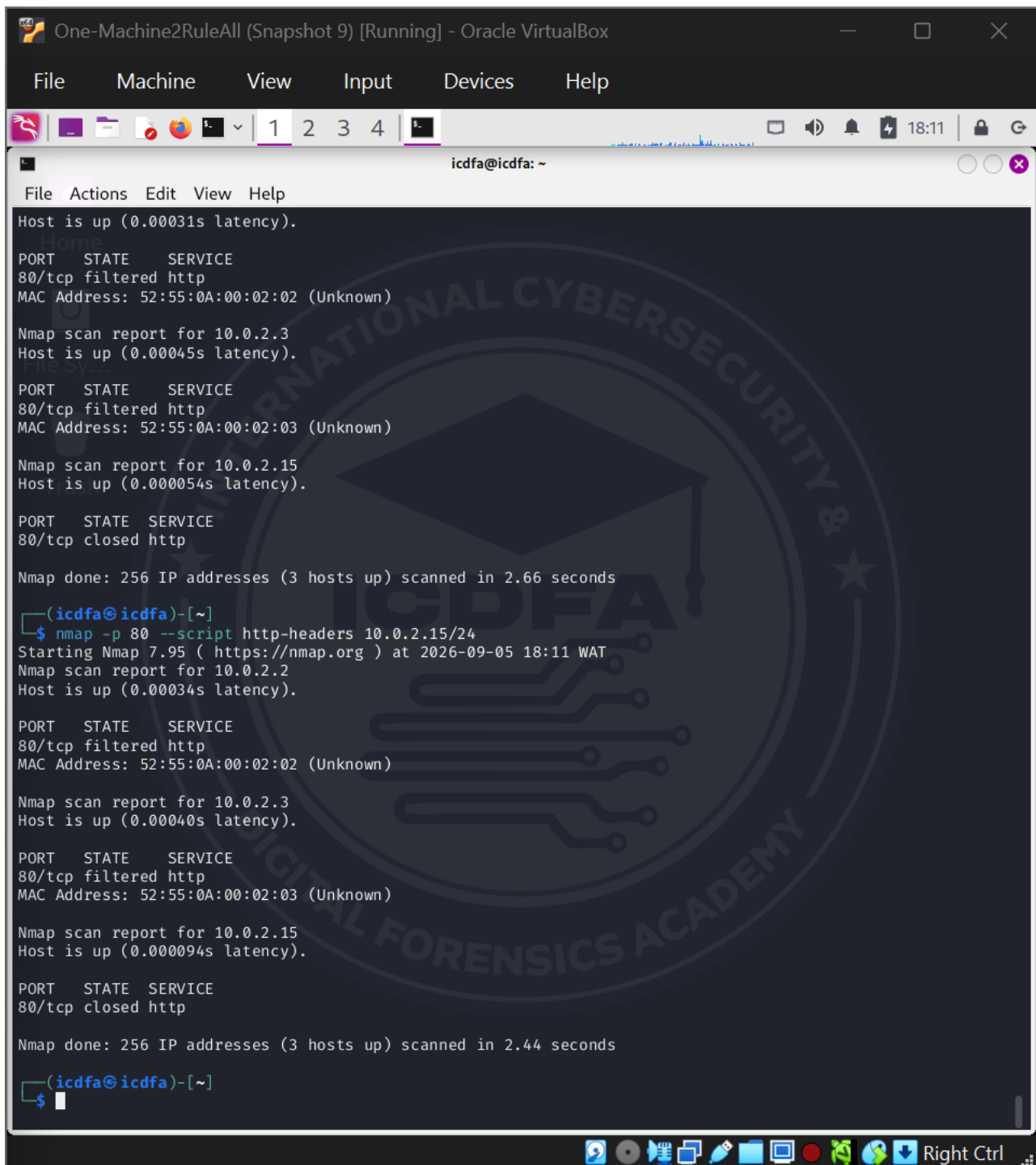
Nmap scan report for 10.0.2.3
Host is up (0.0012s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.51
| dns-nsid:
|_  NSID: internet (696e7465726e6574)
|_  id.server: internet
|_  bind.version: dnsmasq-2.51
MAC Address: 52:55:0A:00:02:03 (Unknown)

Nmap scan report for 10.0.2.15
Host is up (0.0000070s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

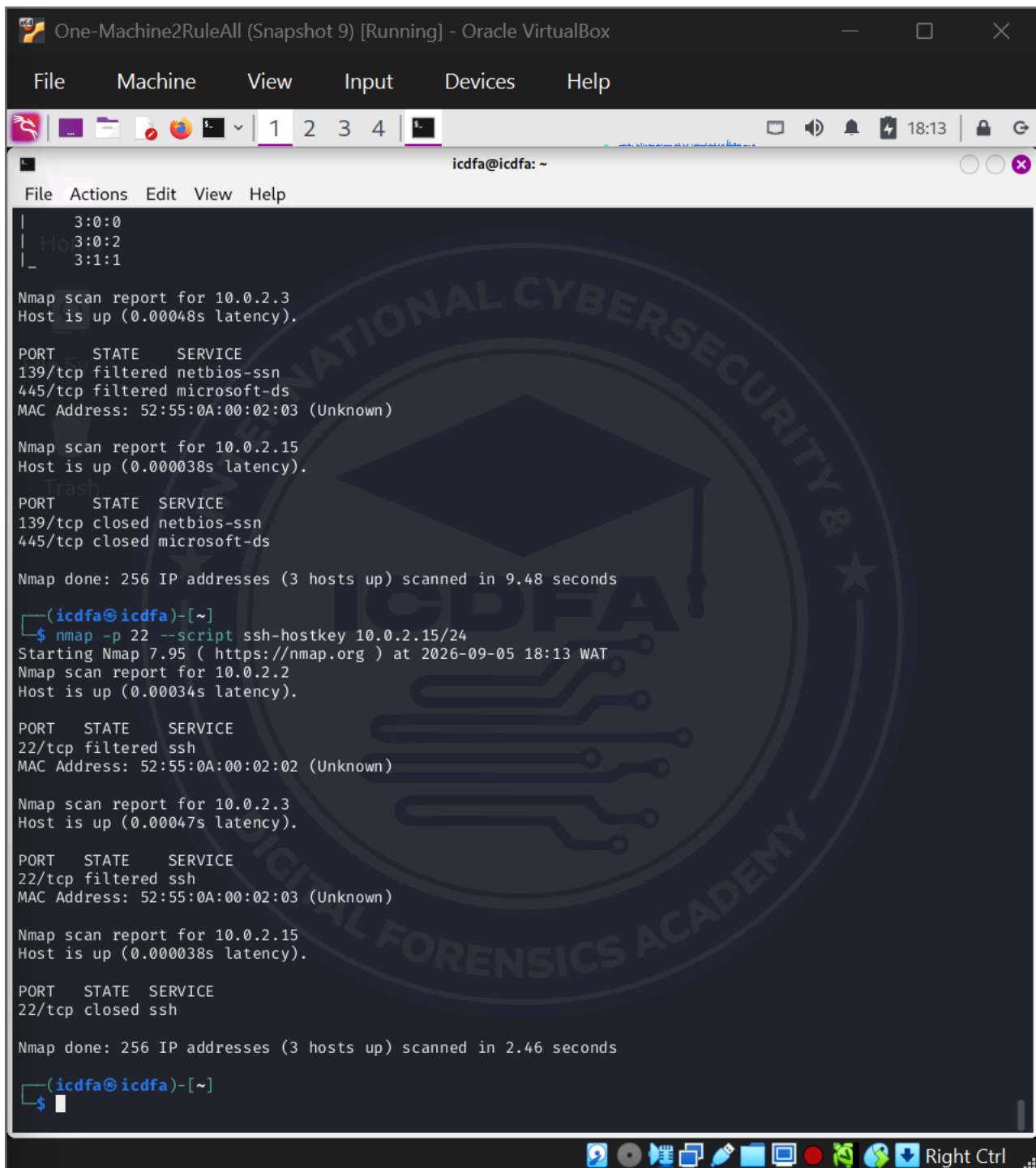
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 27.17 seconds

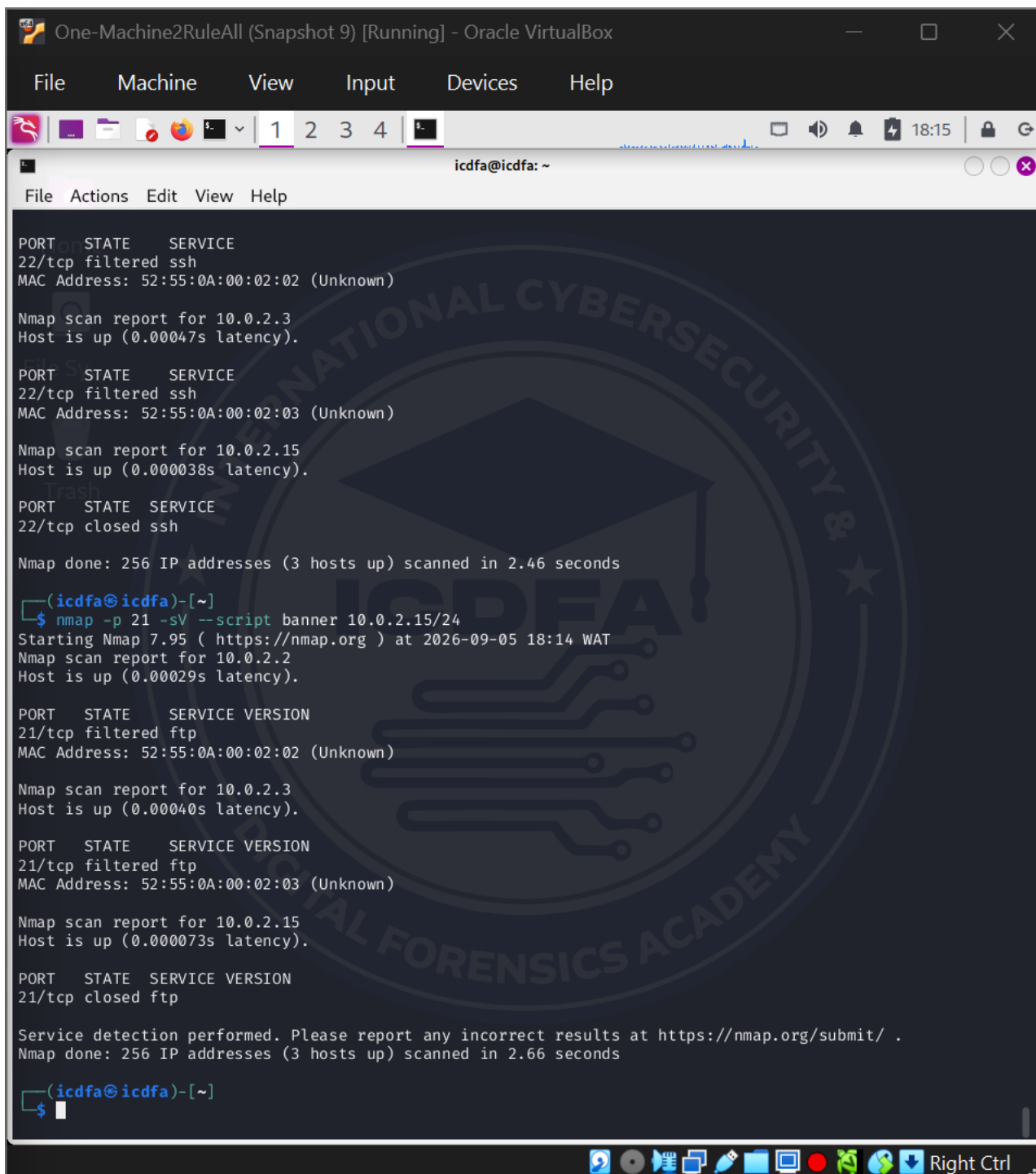
(icdfa@icdfa)-[~]
$
```



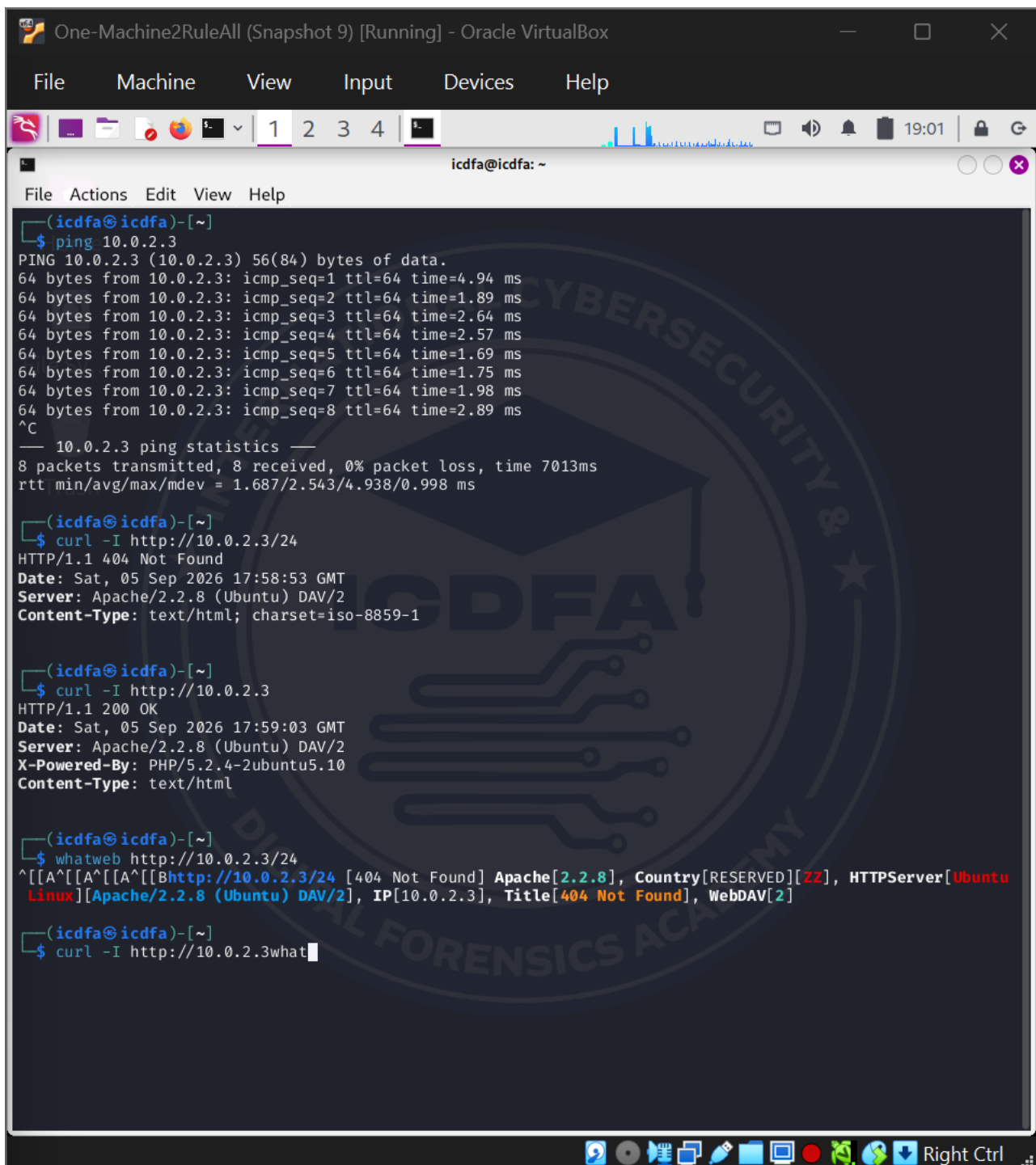








Part 5



```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help

(icdfa@icdfa)-[~]
$ ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=4.94 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=1.89 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=2.64 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=2.57 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=1.69 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=1.75 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=1.98 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=2.89 ms
^C
— 10.0.2.3 ping statistics —
8 packets transmitted, 8 received, 0% packet loss, time 7013ms
rtt min/avg/max/mdev = 1.687/2.543/4.938/0.998 ms

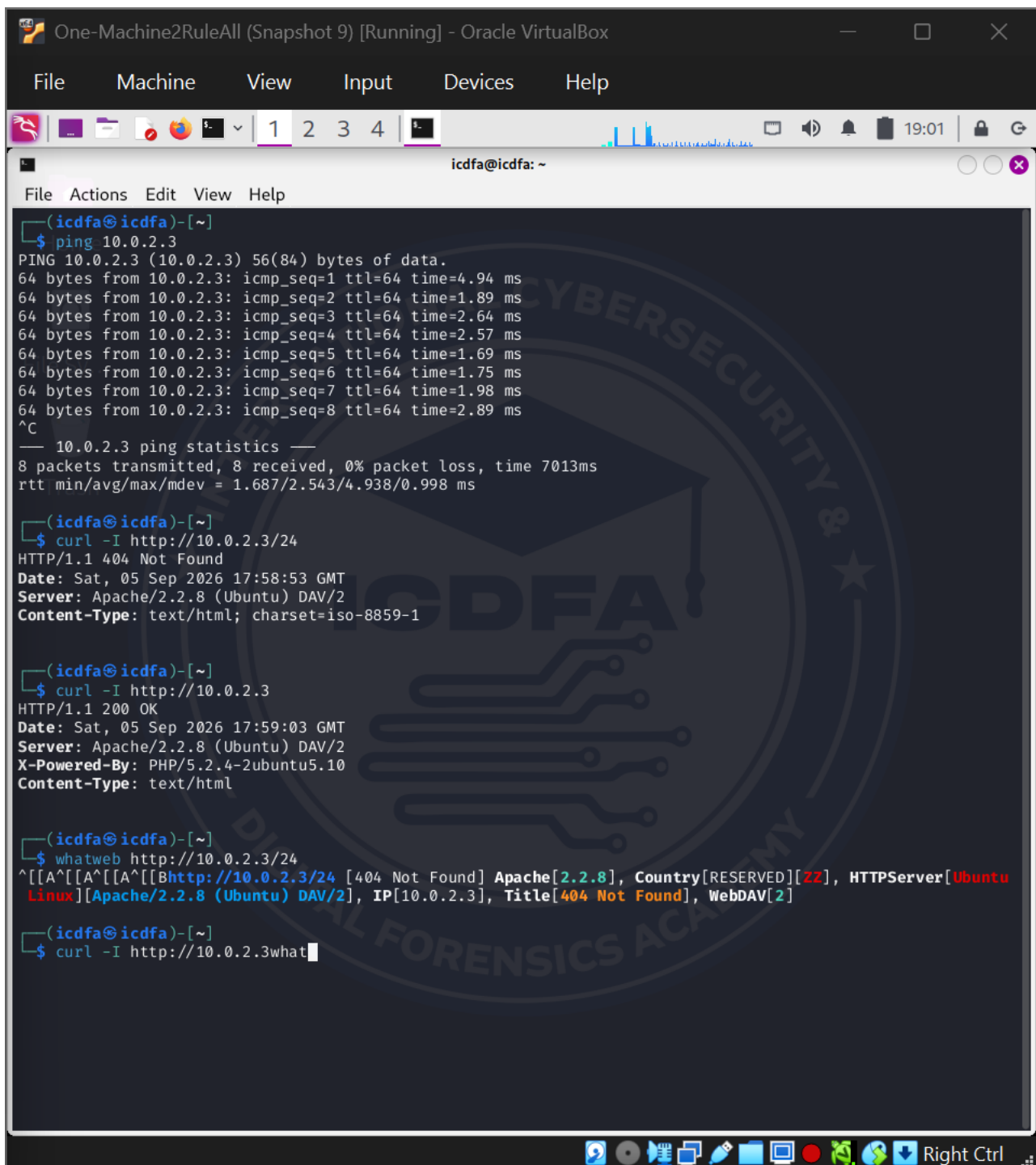
(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3/24
HTTP/1.1 404 Not Found
Date: Sat, 05 Sep 2026 17:58:53 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
Content-Type: text/html; charset=iso-8859-1

(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 17:59:03 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

(icdfa@icdfa)-[~]
$ whatweb http://10.0.2.3/24
^[[A^[[A^[[A^[[Bhttp://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]

(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3what
```

Part 6



```
One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help

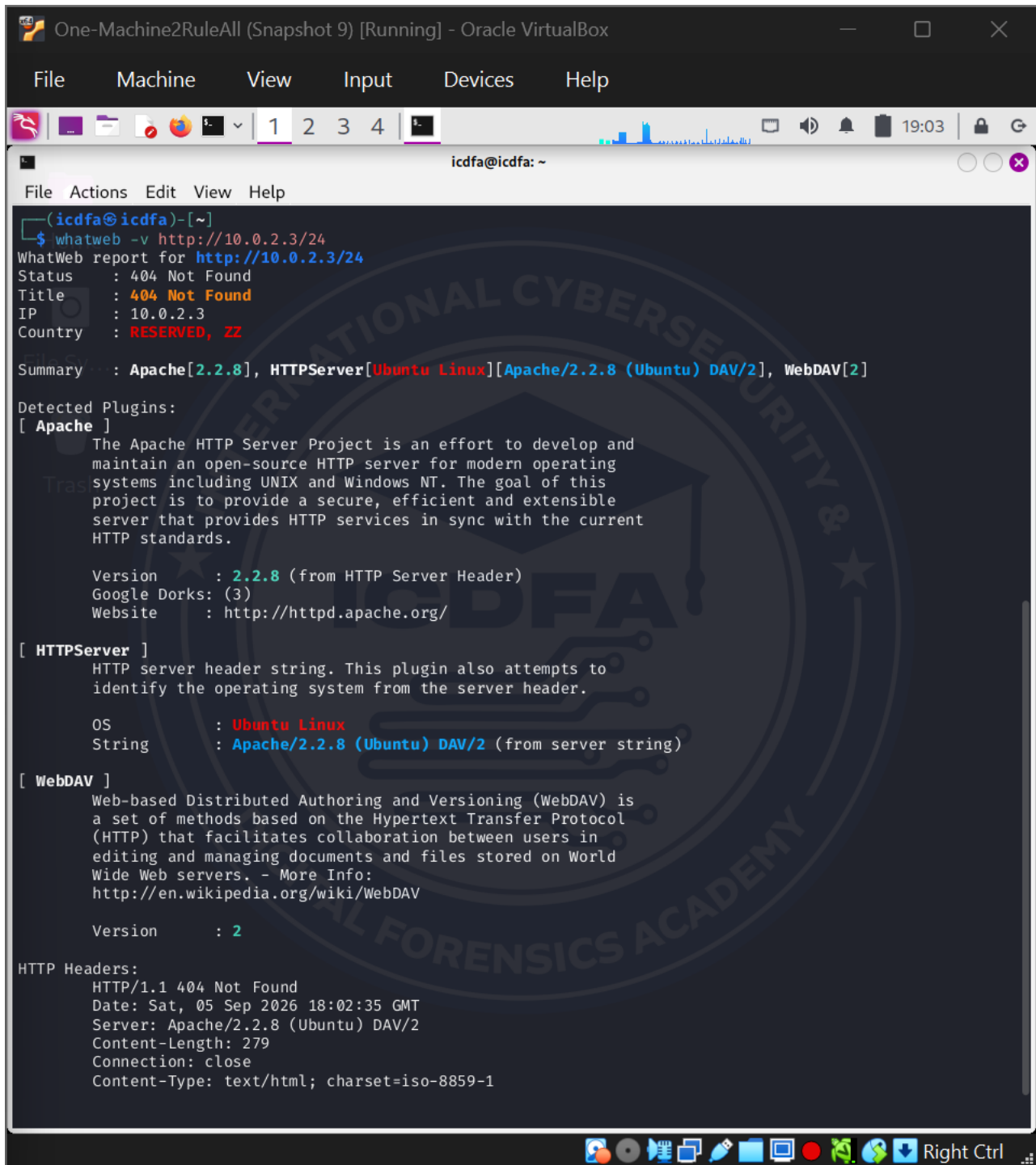
(icdfa@icdfa)-[~]
$ ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=64 time=4.94 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=1.89 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=2.64 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=2.57 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=1.69 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=1.75 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=1.98 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=2.89 ms
^C
— 10.0.2.3 ping statistics —
8 packets transmitted, 8 received, 0% packet loss, time 7013ms
rtt min/avg/max/mdev = 1.687/2.543/4.938/0.998 ms

(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3/24
HTTP/1.1 404 Not Found
Date: Sat, 05 Sep 2026 17:58:53 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
Content-Type: text/html; charset=iso-8859-1

(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 17:59:03 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

(icdfa@icdfa)-[~]
$ whatweb http://10.0.2.3/24
^[[A^[[A^[[Bhttp://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]

(icdfa@icdfa)-[~]
$ curl -I http://10.0.2.3what
```



One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

icdfa@icdfa: ~

File Actions Edit View Help

```
Error in processing commandline options - Agression level must be 1,3, or 4. 1 is invalid.

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3/24
Error in processing commandline options - Agression level must be 1,3, or 4. 1 is invalid.

(icdfa@icdfa)-[~]
$ ping -c 4 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=255 time=3.45 ms

— 10.0.2.3 ping statistics —
4 packets transmitted, 1 received, 75% packet loss, time 3038ms
rtt min/avg/max/mdev = 3.446/3.446/3.446/0.000 ms

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3/24
Error in processing commandline options - Agression level must be 1,3, or 4. 1 is invalid.

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3
Error in processing commandline options - Agression level must be 1,3, or 4. 1 is invalid.

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3/24
^[[A^[AERROR Opening: http://10.0.2.3/24 - execution expired

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3
ERROR Opening: http://10.0.2.3 - execution expired

(icdfa@icdfa)-[~]
$ whatweb -a 3 http://10.0.2.3/24
http://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]

(icdfa@icdfa)-[~]
$ whatweb -a 4 -v http://10.0.2.3/24
^[[D^[[D^[[C^C/usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block (2 levels) in scan': Interrupt
  from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `map'
  from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block in scan'
  from <internal:kernel>:187:in `loop'
  from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:74:in `scan'
  from /usr/bin/whatweb:709:in `<main>'

(icdfa@icdfa)-[~]
$
```

Right Ctrl

One-Machine2RuleAll (Snapshot 9) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

icdfa@icdfa: ~

File Actions Edit View Help

```
— 10.0.2.3 ping statistics —
4 packets transmitted, 1 received, 75% packet loss, time 3038ms
rtt min/avg/max/mdev = 3.446/3.446/3.446/0.000 ms

(icdfa@icdfa)-[~]
$ whatweb -a l http://10.0.2.3/24
Error in processing commandline options - Agression level must be 1,3, or 4. l is invalid.

(icdfa@icdfa)-[~]
$ whatweb -a l http://10.0.2.3
Error in processing commandline options - Agression level must be 1,3, or 4. l is invalid.

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3/24
^[[A^[[AERROR Opening: http://10.0.2.3/24 - execution expired

(icdfa@icdfa)-[~]
$ whatweb -a 1 http://10.0.2.3
ERROR Opening: http://10.0.2.3 - execution expired

(icdfa@icdfa)-[~]
$ whatweb -a 3 http://10.0.2.3/24
http://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]

(icdfa@icdfa)-[~]
$ whatweb -a 4 -v http://10.0.2.3/24
^[[D^[[D^[[C^C/usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block (2 levels) in scan': Interrupt
    from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `map'
    from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block in scan'
    from <internal:kernel>:187:in `loop'
    from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:74:in `scan'
    from /usr/bin/whatweb:709:in `<main>'

(icdfa@icdfa)-[~]
$ whaweb --follow-redirect=always http://10.0.2.3/24
Command 'whaweb' not found, did you mean:
  command 'whatweb' from deb whatweb
Try: sudo apt install <deb name>

(icdfa@icdfa)-[~]
$ whatweb --follow-redirect=always http://10.0.2.3/24
http://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]

(icdfa@icdfa)-[~]
$
```

Right Ctrl

Part 7

Port	Protocol	State	Service	Version/ Evidence	Command
135	TCP	open	msrpc	Microsoft Windows RPC	Nmap -sV 10.0.2.12/24
445	tcp	open	microsoft- ds7		Nmap -sV 10.0.2.12/24
2869	tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)	Nmap -sV 10.0.2.12/24
5995	tcp	open	http	Microsoft HTTPAPI httpd 2.8 (SSDP/UPrP)	Nmap -sV 10.0.2.12/24

Part 8

Q1. What is reconnaissance? The preliminary, non-invasive discovery phase in a security assessment where an analyst systematically gathers information, maps reachable network targets, enumerates exposed ports, and fingerprints running service versions without actively exploiting vulnerabilities.

Q2. What is the difference between host discovery and port scanning? Host discovery (-sn) determines which systems are online and responsive across a subnet without probing service ports. Port scanning actively probes specific transport ports on a discovered host to determine listening daemons.

Q3. Explain open, closed, and filtered port states.

- Open: An application is actively accepting TCP connections or UDP packets.
- Closed: The host is reachable, but no application is listening
- Filtered: A firewall or packet filter drops or intercepts probes, preventing Nmap from determining the state.

Q4. What does -sV do? nmap -sV Enables service version detection by sending application-layer probes to open ports and comparing responses against Nmap's signature database to determine exact software product names and version numbers.

Q5. What does -O do? Enables TCP/IP stack fingerprinting, analyzing subtle implementation differences in TCP/UDP/ICMP packet responses to estimate the target operating system family and kernel version.

Q6. Explain what -A combines and why it is noisier. sudo nmap -A Combines OS detection (-O), version detection (-sV), default NSE scripts (-sC), and traceroute. It generates significantly more active network traffic and probe packets, making it highly conspicuous in firewall and IDS/IPS logs.

Q7. What does -p- mean? Instructs Nmap to scan all 65,535 possible ports instead of its default top 1,000 common port profile.

Q8. Why can UDP scans take longer? UDP is connectionless; open ports rarely return acknowledgments, requiring Nmap to wait for timeout intervals.

Q9. What does -sC do? Runs the default collection of Nmap Scripting Engine (NSE) scripts categorized as safe, non-intrusive enumeration scripts for service discovery and metadata extraction.

Q10. What information did the HTTP title/headers scripts reveal? Extracted the HTML <title>, HTTP response codes (200 OK), and critical server headers such as Server and PHP version signatures.

Q11. What did SMB/SSH/FTP enumeration add? Revealed supported SMB dialect negotiations, host public key algorithms and fingerprint hashes (RSA/DSA), and the raw FTP banner revealing vsftpd 2.3.4.

Q12. What is the difference between Nmap service detection and WhatWeb fingerprinting? Nmap focuses on network/transport-layer service binding and daemon versions across all ports. WhatWeb is a specialized Layer 7 (HTTP) web technology scanner that identifies CMS platforms, JavaScript frameworks, web server modules, cookies, and embedded technologies.

Q13. How did WhatWeb levels 1, 3, and 4 differ?

- Level 1 (Stealthy): Single GET request; passive analysis of headers and page body.
- Level 3 (Aggressive): Issues additional focused requests to probe specific assets and paths.
- Level 4 (Heavy): Recursive exhaustive probing against all plugin signatures, producing high request volume.

Q14. Why must aggressive scanning remain inside the authorised lab?

Aggressive scans generate large traffic volumes that can crash legacy/fragile network daemons, trigger security defense systems, and violate computer misuse laws and enterprise acceptable-use policies on unauthorized networks.